

Stat 371 Practice Problem Session #3

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Overview of content covered:

1. Probability & Events
2. Expectation & Variance
3. Random Variables (Binomial) & pbinom/dbinom functions
4. Conditional Probability

1. Coin Flips

You flip a biased coin 4 times and count the number of heads, calling it N . The biased coin has a 45% chance of comes up tails

a. What's the underlying sample space? (Note: depending on how you interpret this there could be multiple answers)

Ans: The possible length 4 sequences of Heads/Tails (Eg TTTT, HHHH, etc)

b. What type of random variable is the number of heads? What are its parameters?

Ans: Binomial($n = 4$, $p = 0.45$)

c. Fill in the following table

Number of Heads	Probability
0	$.45^4$
1	$4 \cdot 0.45^3 \cdot 0.55$
2	$6 \cdot 0.45^2 \cdot 0.55^2$
3	$4 \cdot 0.45^1 \cdot 0.55^3$
4	0.55^4

- Compute it by hand first
- Check your work with dbinom() function in R (Hint: dbinom(k,n,p) gives you

the probability for exactly k successes of n trials with probability p)

Here is an example of checking one entry:

```
In [8]: dbinom(3, 4, 0.55)
```

0.299475

```
In [7]: 4 * 0.45 * 0.55^3
```

0.299475

d. Compute the expected number of heads $E[N]$, the variance $Var(N)$, the standard deviation $sd(N)$, and $E[N^2] - (E[N])^2$. Notice anything similar between these quantities?

$$\begin{aligned} E[N] &= P * N = \\ &= .45^4 * 0 + 4 * .45^3 * 0.55 * 1 + 6 * .45^2 * 0.55^2 * 2 + 4 * 0.45 * 0.55^3 * 3 + 0.55^4 * 4 \end{aligned}$$

$$\begin{aligned} E[N^2] &= P(N = 0) * 0 + P(N = 1) * 1 + P(N = 2) * 4 + P(N = 3) * 9 + P(N = 4) * 16 \\ &= .45^4 * 0 + 4 * .45^3 * 0.55 * 1 + 6 * .45^2 * 0.55^2 * 4 + 4 * 0.45 * 0.55^3 * 9 + 0.55^4 * 16 \\ &= 5.38 \end{aligned}$$

$$Var[N] = P * (1 - P)N = 0.55 * 0.45 * 4 = 0.99$$

Notice that $E[N^2] - (E[N])^2 = 0.99 = Var(N)$. This is a nice way to compute variance.

e. What's the probability of the event that the coin comes up heads more than once?

- Compute this by hand using the table
- Check your work using the `pbinom()` function in R (Hint: `pbinom(k,n,p)` gives you the probability for AT MOST k successes of n trials with probability p)

$$P(N > 1) = P(N = 2) + P(N = 3) + P(N = 4) = 6 * 0.45^2 * 0.55^2 + 4 * 0.45 * 0.55^3 + 0.55^4$$

```
In [15]: 1 - pbinom(1, 4, 0.55)
```

0.75851875

f. What's the probability of the event that the coin comes up heads an even number of times?

$$P(N_{even}) = P(N = 0) + P(N = 2) + P(N = 4) = 0.45^4 + 6 * 0.45^2 * 0.55^2 + 0.55^4$$

g. What's the probability that the number of heads is 4 given that the first coin

flip was heads? What about if it was tails?

$$P(N = 4 | \text{first flip heads}) = \frac{P(N = 4 \text{ and first flip heads})}{P(\text{first flip heads})} = \frac{0.55^4}{0.55} = 0.55^3 =$$

where we use the fact that the event where $N = 4$ requires that the first flip is heads, so $P(N = 4 \text{ and first flip heads}) = P(N = 4)$

2. Birthday Problem

Suppose you have a group of N people. We say there is a shared birthday if at least two people in the group have the same birthday.

a. Explain why the probability of this event is:

$$P(\text{shared birthday}) = 1 - \frac{365 * (365 - 1) * (365 - 2) * \dots * (365 - N + 1)}{365^N}$$

Ans: This is just computing the complement of the event of no shared birthday. We can then just compute the number of no shared by birthdays by counting the number of ways N people can have no birthdays and dividing by the total number of ways N people can have any birthdays.

b. Compute the probability of a shared birthday for a group of $N = 5$ people.

Ans:

$$P(\text{shared birthday}) = 1 - \frac{(365 - 1) * (365 - 2) * \dots * (365 - N + 1)}{365^{N-1}} = 1 - \frac{3}{365}$$